

Introduction to Data Assimilation

Lecture 4 Supplementary

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Abstract

This note documents the Kalman filter in 1D.

Keywords Kalman filter

1 Kalman Filter: 1D Example [1]

Real-time prediction with inaccurate or intermittent measurements requires combining prior (model) forecasts with observations. In this chapter we derive the closed-form one-dimensional Kalman filter as the Bayesian analysis step for a simplified yet instructive setting and outline its use for a complex Ornstein–Uhlenbeck (OU) process. We also state stability considerations and the role of model/observation error covariances.

1.1 Overview and Notation

We work with complex-valued states and noises while the observation operator is real (as in the reference). Let $u_m \in \mathbb{C}$ denote the true state at time index $m \in \mathbb{Z}_{\geq 0}$ and let $v_m \in \mathbb{C}$ be the observation. We adopt the forecast/analysis notation

$\bar{u}_{m|m} \in \mathbb{C}$ (posterior/analysis mean at time m), $\bar{u}_{m+1|m} \in \mathbb{C}$ (prior/forecast mean at $m+1$),

with corresponding scalar error covariances

$$r_{m|m} = \mathbb{E}[(u_m - \bar{u}_{m|m})(u_m - \bar{u}_{m|m})^*] \in \mathbb{R}_{\geq 0}, \quad r_{m+1|m} = \mathbb{E}[(u_{m+1} - \bar{u}_{m+1|m})(u_{m+1} - \bar{u}_{m+1|m})^*].$$

1.2 Stochastic Model

Forecast (state) model. The complex linear dynamics with complex Gaussian system noise is

$$u_{m+1} = F u_m + \sigma_{m+1}, \quad F \in \mathbb{C}, \quad \sigma_{m+1} \sim \mathcal{N}(0, r), \quad r = \mathbb{E}[\sigma_{m+1} \sigma_{m+1}^*] > 0. \quad (1.1)$$

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Herein, the system noise is understood as

$$\sigma_{m+1} = \frac{\sigma_{1,m+1} + i \sigma_{2,m+1}}{\sqrt{2}},$$

where $\sigma_{j,m+1} \in \mathbb{R}$ are independent, unbiased Gaussians with variances $\langle \sigma_{j,m+1} \sigma_{j,m+1}^* \rangle$ ($j = 1, 2$), so that

$$r := \mathbb{E}[\sigma_{m+1} \sigma_{m+1}^*] := \langle \sigma_{m+1} \sigma_{m+1}^* \rangle = \frac{1}{2} \sum_{j=1}^2 \langle \sigma_{j,m+1} \sigma_{j,m+1}^* \rangle > 0. \quad (1.2)$$

Observation model. A linear real observation operator $g \in \mathbb{R}$ with complex Gaussian observation noise:

$$v_{m+1} = g u_{m+1} + \sigma_{m+1}^o, \quad \sigma_{m+1}^o \sim \mathcal{N}(0, r^o), \quad r^o = \mathbb{E}[\sigma_{m+1}^o (\sigma_{m+1}^o)^*] > 0, \quad (1.3)$$

independent of σ_{m+1} and of past states/noises.

The likelihood at t_{m+1} is Gaussian:

$$p(v_{m+1} \mid u_{m+1}) \sim \mathcal{N}(g u_{m+1}, r^o).$$

1.3 Complex Gaussian Basics (Scalar)

A scalar complex normal random variable $z \in \mathbb{C}$ with mean μ and (Hermitian) variance $s > 0$ is written as $z \sim \mathcal{N}(\mu, s)$ with density

$$p(z) = \frac{1}{\pi s} \exp\left(-\frac{|z - \mu|^2}{s}\right).$$

Independence of complex Gaussian variables is characterized by vanishing cross-covariances, e.g., $\mathbb{E}[\sigma_{m+1} (\sigma_{m+1}^o)^*] = 0$.

1.4 Forecast (Prediction) Step

Let $e_{m|m} := u_m - \bar{u}_{m|m}$ and $e_{m+1|m} := u_{m+1} - \bar{u}_{m+1|m}$ and denote the prior/forecast mean and covariance by

$$\bar{u}_{m+1|m} := \mathbb{E}[u_{m+1} \mid v_{1:m}], \quad r_{m+1|m} := \mathbb{E}[(u_{m+1} - \bar{u}_{m+1|m})(u_{m+1} - \bar{u}_{m+1|m})^*],$$

and the posterior/analysis mean and covariance by $\bar{u}_{m|m}$ and $r_{m|m}$ respectively. Propagating (1.1) gives the forecast formulas

$$\bar{u}_{m+1|m} = F \bar{u}_{m|m}, \quad r_{m+1|m} = F r_{m|m} F^* + r. \quad (1.4)$$

Forecast (Prediction) covariance derivation. From the state model

$$u_{m+1} = F u_m + \sigma_{m+1}, \quad \mathbb{E}[\sigma_{m+1}] = 0, \quad r := \mathbb{E}[\sigma_{m+1}\sigma_{m+1}^*] > 0,$$

and the forecast mean $\bar{u}_{m+1|m} = F \bar{u}_{m|m}$, define the prior error

$$e_{m+1|m} := u_{m+1} - \bar{u}_{m+1|m}.$$

Using $\bar{u}_{m+1|m} = F \bar{u}_{m|m}$ and $e_{m|m} := u_m - \bar{u}_{m|m}$, we have

$$e_{m+1|m} = (F u_m + \sigma_{m+1}) - F \bar{u}_{m|m} = F(u_m - \bar{u}_{m|m}) + \sigma_{m+1} = F e_{m|m} + \sigma_{m+1}.$$

Hence the prior covariance is

$$\begin{aligned} r_{m+1|m} &:= \mathbb{E}[e_{m+1|m} e_{m+1|m}^*] = \mathbb{E}[(F e_{m|m} + \sigma_{m+1})(F e_{m|m} + \sigma_{m+1})^*] \\ &= \mathbb{E}[F e_{m|m} e_{m|m}^* F^*] + \mathbb{E}[F e_{m|m} \sigma_{m+1}^*] + \mathbb{E}[\sigma_{m+1} e_{m|m}^* F^*] + \mathbb{E}[\sigma_{m+1} \sigma_{m+1}^*]. \end{aligned}$$

By independence of σ_{m+1} from $e_{m|m}$ and $\mathbb{E}[\sigma_{m+1}] = 0$, the cross terms vanish:

$$\mathbb{E}[F e_{m|m} \sigma_{m+1}^*] = F \mathbb{E}[e_{m|m}] \mathbb{E}[\sigma_{m+1}^*] = 0, \quad \mathbb{E}[\sigma_{m+1} e_{m|m}^* F^*] = \mathbb{E}[\sigma_{m+1}] \mathbb{E}[e_{m|m}^*] F^* = 0.$$

Therefore,

$$r_{m+1|m} = F \mathbb{E}[e_{m|m} e_{m|m}^*] F^* + \mathbb{E}[\sigma_{m+1} \sigma_{m+1}^*] = F r_{m|m} F^* + r,$$

which is the desired forecast (prior) covariance update.

1.5 Analysis (Filtering) Step: Two Derivations of the Kalman Gain

Form of the affine update. Given v_{m+1} , we update the mean with an affine rule

$$\bar{u}_{m+1|m+1} = \bar{u}_{m+1|m} + K_{m+1}(v_{m+1} - g \bar{u}_{m+1|m}), \quad K_{m+1} \in \mathbb{R}, \quad (1.5)$$

and define the posterior covariance

$$r_{m+1|m+1} = \mathbb{E}[(u_{m+1} - \bar{u}_{m+1|m+1})(u_{m+1} - \bar{u}_{m+1|m+1})^*]. \quad (1.6)$$

Herein, K_{m+1} is called the Kalman gain, and $v_{m+1} - g \bar{u}_{m+1|m}$ is called the innovation.

1.5.1 (A) Bayesian/Conditional-Density Derivation via Minimizing $J(u)$

Bayes' rule (density form). Let u be the (latent) state and v the observation. Assume joint density $p(u, v)$ with marginals $p(u)$ and $p(v)$ and conditional densities

$$p(u | v) = \frac{p(u, v)}{p(v)} \text{ and } p(v | u) = \frac{p(u, v)}{p(u)}.$$
 Then

$$p(u | v) = \frac{p(u, v)}{p(v)} = \frac{p(v | u) p(u)}{p(v)} \propto p(v | u) p(u),$$

where the proportionality constant

$$p(v) = \int p(v | u) p(u) du$$

depends only on v (the data) and not on u (the state). Thus, the posterior density is proportional to the likelihood times the prior:

$$\boxed{p(u | v) = \frac{1}{Z(v)} p(v | u) p(u), \quad Z(v) = \int p(v | u) p(u) du.}$$

Kalmain gain derivation. The likelihood and prior at time $m+1$ are Gaussian:

$$p(v_{m+1} | u_{m+1}) \propto \exp\left(-\frac{|v_{m+1} - gu_{m+1}|^2}{r^o}\right), \quad p(u_{m+1}) \propto \exp\left(-\frac{|u_{m+1} - \bar{u}_{m+1|m}|^2}{r_{m+1|m}}\right).$$

Hence the posterior density is proportional to $\exp(-\frac{1}{2}J(u))$ with

$$J(u) = \frac{|u - \bar{u}_{m+1|m}|^2}{r_{m+1|m}} + \frac{|v_{m+1} - gu|^2}{r^o}.$$

Since J is real-valued and strictly convex in $u \in \mathbb{C}$, the minimizer satisfies the first-order optimality condition (using, e.g., Wirtinger calculus $\partial J / \partial u^* = 0$):

$$\frac{\partial J}{\partial u^*} = \frac{u - \bar{u}_{m+1|m}}{r_{m+1|m}} - \frac{g(v_{m+1} - gu)}{r^o} = 0.$$

Rearranging,

$$\left(\frac{1}{r_{m+1|m}} + \frac{g^2}{r^o}\right)u = \frac{\bar{u}_{m+1|m}}{r_{m+1|m}} + \frac{g v_{m+1}}{r^o}.$$

Solving for u and comparing with (1.5) yields the Kalman gain

$$\boxed{K_{m+1} = \frac{g r_{m+1|m}}{r^o + g^2 r_{m+1|m}} \in [0, 1/g]} \quad (1.7)$$

and the posterior mean update (1.5).

1.5.2 (B) Least-Squares (Minimum-Error-Variance) Derivation

We keep the notation: the prior (forecast) error at time $m+1$ is

$$e_{m+1|m} := u_{m+1} - \bar{u}_{m+1|m}, \quad \mathbb{E}[e_{m+1|m}] = 0, \quad r_{m+1|m} := \mathbb{E}[e_{m+1|m} e_{m+1|m}^*] \in \mathbb{R}_{\geq 0},$$

and the observation model is $v_{m+1} = g u_{m+1} + \sigma_{m+1}^o$ with $g \in \mathbb{C}$ (real g is a special case), independent, zero-mean complex Gaussian noise σ_{m+1}^o , and variance $r^o := \mathbb{E}[\sigma_{m+1}^o (\sigma_{m+1}^o)^*] > 0$. Define the innovation

$$d_{m+1} := v_{m+1} - g \bar{u}_{m+1|m} = g e_{m+1|m} + \sigma_{m+1}^o,$$

and consider the affine estimator

$$\bar{u}_{m+1|m+1} = \bar{u}_{m+1|m} + K_{m+1} d_{m+1}, \quad K_{m+1} \in \mathbb{C}.$$

The posterior error is

$$e_{m+1|m+1} := u_{m+1} - \bar{u}_{m+1|m+1} = (1 - K_{m+1}g) e_{m+1|m} - K_{m+1} \sigma_{m+1}^o.$$

We minimize the mean-square error

$$\phi(K) := \mathbb{E}[|e_{m+1|m+1}|^2] = \mathbb{E}[(1 - Kg)e - K\sigma^o][(1 - Kg)e - K\sigma^o]^*.$$

Independence and zero means imply $\mathbb{E}[e(\sigma^o)^*] = 0$, so the cross terms vanish and

$$\phi(K) = |1 - Kg|^2 r_{m+1|m} + |K|^2 r^o.$$

Treat K and K^* as independent (Wirtinger calculus) and set the stationary condition

$$\frac{\partial \phi}{\partial K^*} = 0.$$

Compute

$$\frac{\partial}{\partial K^*}(|1 - Kg|^2) = \frac{\partial}{\partial K^*}((1 - Kg)(1 - Kg)^*) = (1 - Kg) \frac{\partial}{\partial K^*}(1 - Kg)^* = -(1 - Kg)g^*,$$

and $\frac{\partial}{\partial K^*}(|K|^2) = \frac{\partial}{\partial K^*}(KK^*) = K$. Hence

$$\frac{\partial \phi}{\partial K^*} = r_{m+1|m}(-g^*(1 - Kg)) + r^o K = -r_{m+1|m}g^* + r_{m+1|m}|g|^2 K + r^o K.$$

Setting this to zero yields

$$(|g|^2 r_{m+1|m} + r^o) K = r_{m+1|m} g^*,$$

and therefore the complex scalar Kalman gain

$$K_{m+1} = \frac{r_{m+1|m} g^*}{r^o + |g|^2 r_{m+1|m}} \in \mathbb{C}.$$

If $g \in \mathbb{R}$, this reduces to $K_{m+1} = \frac{g r_{m+1|m}}{r^o + g^2 r_{m+1|m}}$.

Posterior covariance check (Joseph form). With the optimal K_{m+1} ,

$$r_{m+1|m+1} = \mathbb{E}[|e_{m+1|m+1}|^2] = (1 - K_{m+1}g)(1 - K_{m+1}g)^* r_{m+1|m} + |K_{m+1}|^2 r^o = \frac{r_{m+1|m} r^o}{r^o + |g|^2 r_{m+1|m}}.$$

Posterior Covariance Update (complex scalar) and Joseph Form

Let the prior (forecast) error and covariance at time $m+1$ be

$$e_{m+1|m} := u_{m+1} - \bar{u}_{m+1|m}, \quad r_{m+1|m} := \mathbb{E}[e_{m+1|m} e_{m+1|m}^*] \in \mathbb{R}_{\geq 0}.$$

With the innovation $d_{m+1} = v_{m+1} - g \bar{u}_{m+1|m} = g e_{m+1|m} + \sigma_{m+1}^o$ and the affine analysis

$$\bar{u}_{m+1|m+1} = \bar{u}_{m+1|m} + K_{m+1} d_{m+1}, \quad K_{m+1} \in \mathbb{C},$$

the posterior (analysis) error is

$$e_{m+1|m+1} := u_{m+1} - \bar{u}_{m+1|m+1} = (1 - K_{m+1}g) e_{m+1|m} - K_{m+1} \sigma_{m+1}^o.$$

Independence and zero means give $\mathbb{E}[e_{m+1|m}(\sigma_{m+1}^o)^*] = 0$. Thus

$$\begin{aligned} r_{m+1|m+1} &:= \mathbb{E}[e_{m+1|m+1} e_{m+1|m+1}^*] \\ &= \mathbb{E}[(1 - Kg)e - K\sigma^o][(1 - Kg)e - K\sigma^o]^* \\ &= (1 - K_{m+1}g) r_{m+1|m} (1 - K_{m+1}g)^* + K_{m+1} r^o K_{m+1}^* \\ &= (1 - Kg)(1 - Kg)^* r_{m+1|m} + |K|^2 r^o \quad (\text{assuming real}) \\ &= (1 - K_{m+1}g) r_{m+1|m} \quad (\text{after substituting } K_{m+1} \text{ from (1.7)}) \end{aligned} \tag{1.8}$$

Equation (1.8) is the *Joseph form* in the complex scalar case; it holds for any (possibly suboptimal) K and guarantees $r_{m+1|m+1} \geq 0$.

Equivalence of the Joseph Form and the Simplified Posterior-Covariance Update (1D KF, Complex Scalar)

Setup and notation. Let $u \in \mathbb{C}$ be the state, $v \in \mathbb{C}$ the observation, $g \in \mathbb{C}$ the (scalar) observation operator, and let $r := r_{m+1|m} \in \mathbb{R}_{\geq 0}$ be the prior (forecast) variance at time $m+1$. The observation-noise variance is $r^o > 0$. For a complex scalar gain $K \in \mathbb{C}$, the *Joseph form* of the posterior variance is

$$r_{m+1|m+1} = (1 - Kg)(1 - Kg)^* r + |K|^2 r^o. \tag{1.9}$$

The *simplified form* is

$$r_{m+1|m+1} = (1 - Kg) r, \tag{1.10}$$

which is valid *at the optimal Kalman gain*. The complex scalar Kalman gain is

$$K^* = \frac{r g^*}{r^o + |g|^2 r}, \quad D := r^o + |g|^2 r > 0. \tag{1.11}$$

Theorem 1.1 (Equivalence at the optimal gain). *For the one-dimensional complex scalar Kalman filter, if $K = K^*$ in (1.11), then the Joseph form (1.9) and the simplified form (1.10) coincide:*

$$(1 - K^*g)(1 - K^*g)^* r + |K^*|^2 r^o = (1 - K^*g) r = \frac{r r^o}{r^o + |g|^2 r}.$$

Proof. Insert $K^\star$ from (1.11). Then

$$K^\star g = \frac{r g^\star}{D} g = \frac{|g|^2 r}{D} \in \mathbb{R}_{\geq 0}, \quad |K^\star|^2 = \frac{r^2 |g|^2}{D^2}.$$

Hence

$$(1 - K^\star g)(1 - K^\star g)^\star = (1 - K^\star g)^2 = \left(1 - \frac{|g|^2 r}{D}\right)^2 = 1 - 2\frac{|g|^2 r}{D} + \frac{|g|^4 r^2}{D^2}.$$

Using these identities in the Joseph form (1.9) gives

$$\begin{aligned} r_{m+1|m+1} &= \left(1 - 2\frac{|g|^2 r}{D} + \frac{|g|^4 r^2}{D^2}\right)r + \frac{r^2 |g|^2}{D^2} r^o \\ &= r - 2\frac{|g|^2 r^2}{D} + \frac{|g|^4 r^3}{D^2} + \frac{|g|^2 r^2 r^o}{D^2}. \end{aligned}$$

Factor the last two terms:

$$\frac{|g|^4 r^3}{D^2} + \frac{|g|^2 r^2 r^o}{D^2} = \frac{|g|^2 r^2}{D^2} (|g|^2 r + r^o) = \frac{|g|^2 r^2}{D^2} D = \frac{|g|^2 r^2}{D}.$$

Therefore

$$r_{m+1|m+1} = r - 2\frac{|g|^2 r^2}{D} + \frac{|g|^2 r^2}{D} = r - \frac{|g|^2 r^2}{D} = r\left(1 - \frac{|g|^2 r}{D}\right) = r \frac{r^o}{D} = \frac{r r^o}{r^o + |g|^2 r}.$$

But with $K^\star g = \frac{|g|^2 r}{D}$, the simplified form (1.10) gives

$$(1 - K^\star g) r = \left(1 - \frac{|g|^2 r}{D}\right)r = \frac{r r^o}{D}.$$

Thus the two expressions are equal at $K = K^\star$. □

Corollary 1.2 (Equivalent characterizations). *The following are equivalent for the 1D complex scalar KF:*

1. $K = K^\star$ in (1.11) (Kalman gain solves the normal equation $(|g|^2 r + r^o)K = g^\star r$).
2. The posterior covariance equals the simplified form (1.10).
3. The Joseph form (1.9) reduces to the simplified value $r r^o / (r^o + |g|^2 r)$.

Aspect	Joseph form	Simplified form
Exact equality	Equals simplified only at $K = K^*$	Valid only at K^*
Inputs needed	Uses r, r^o, g, K	Uses r, g, K
PSD guarantee	Always $r_+ \geq 0$ for any K	Not guaranteed for arbitrary K
Numerical stability	More robust to roundoff/approx. K	More compact; may lose PSD
Closed form at K^*	$r_+ = \frac{r r^o}{r^o + g ^2 r}$	Same value at K^*
Cost (1D)	Slightly higher (extra multiplies)	Minimal
Matrix generalization	$(I - KH)P(I - KH)^* + KRK^*$	$(I - KH)P$ (valid at K^*)

Table 1: Posterior-variance updates in 1D complex-valued Kalman filtering. Here $r_+ := r_{m+1|m+1}$.

Closed form with the optimal Kalman gain. For the complex scalar setting, the optimal gain is

$$K_{m+1} = \frac{r_{m+1|m} g^*}{r^o + |g|^2 r_{m+1|m}}.$$

Let $D := r^o + |g|^2 r_{m+1|m}$. Then

$$Kg = \frac{r_{m+1|m} |g|^2}{D} \in \mathbb{R}_{\geq 0}, \quad |K|^2 = \frac{r_{m+1|m}^2 |g|^2}{D^2}.$$

Insert these into (1.8):

$$\begin{aligned}
r_{m+1|m+1} &= \left(1 - \frac{r_{m+1|m} |g|^2}{D}\right)^2 r_{m+1|m} + \frac{r_{m+1|m}^2 |g|^2}{D^2} r^o \\
&= r_{m+1|m} - 2 \frac{r_{m+1|m}^2 |g|^2}{D} + \frac{r_{m+1|m}^3 |g|^4}{D^2} + \frac{r_{m+1|m}^2 |g|^2 r^o}{D^2} \\
&= r_{m+1|m} - \frac{r_{m+1|m}^2 |g|^2}{D} = \frac{r_{m+1|m} r^o}{r^o + |g|^2 r_{m+1|m}}.
\end{aligned}$$

Equivalently, since $1 - Kg = \frac{r^o}{D}$,

$$r_{m+1|m+1} = (1 - K_{m+1}g) r_{m+1|m} = \frac{r_{m+1|m} r^o}{r^o + |g|^2 r_{m+1|m}}.$$

The first equality highlights the contraction of the prior covariance by the factor $(1 - Kg)$; the second gives the explicit posterior variance in terms of $r_{m+1|m}$, r^o , and g .

1.6 Algorithmic Summary (1D Complex Kalman Filter)

Given $(\bar{u}_{m|m}, r_{m|m})$:

$$\begin{aligned} \text{Forecast: } \bar{u}_{m+1|m} &= F \bar{u}_{m|m}, & r_{m+1|m} &= F r_{m|m} F^* + r. \\ \text{Analysis: } K_{m+1} &= \frac{g r_{m+1|m}}{r^o + g^2 r_{m+1|m}}, \\ \bar{u}_{m+1|m+1} &= \bar{u}_{m+1|m} + K_{m+1} (v_{m+1} - g \bar{u}_{m+1|m}), \\ r_{m+1|m+1} &= (1 - K_{m+1} g) r_{m+1|m}. \end{aligned}$$

1.7 Remarks on Complex Setting

All updates are scalar and maintain the covariance real and nonnegative. Complex arithmetic enters through $F \in \mathbb{C}$ and the noises; the observation operator $g \in \mathbb{R}$ and the covariances $r, r^o, r_{(\cdot)}$ are real and nonnegative. The derivations above are unchanged from the real case after replacing transposes by conjugate-transposes and using the scalar identity $|z|^2 = z z^*$ for $z \in \mathbb{C}$.

2 Complex OU example (for simulation)

Consider the complex OU SDE

$$du(t) = (-\gamma + i\omega)u(t) dt + \sigma dW(t), \quad \gamma, \sigma > 0, \omega \in \mathbb{R}, \quad (2.1)$$

with complex white noise

$$dW(t) = \frac{dW_1(t) + i dW_2(t)}{\sqrt{2}},$$

where W_j are independent Wiener processes; formally $\dot{W}_j(t)$ are zero-mean white noises with

$$\langle \dot{W}_j(t) \rangle = 0, \quad \langle \dot{W}_j(t) \dot{W}_j(s) \rangle = \delta(t-s), \quad \langle \dot{W}_i(t) \dot{W}_j(s) \rangle = 0 \quad (i \neq j).$$

Time-discretizing this SDE (e.g. Euler–Maruyama) and using (1.4) yields the standard 1D complex Kalman filter testbed.

Remarks.

- In the scalar complex setting, all covariances $(r, r^o, r_{(\cdot)})$ are real and nonnegative; conjugation appears only in inner-product forms $|z|^2 = z z^*$.
- The analysis step is optimal in the least-squares sense under linear/Gaussian assumptions and coincides with the Bayesian posterior mean.

The Lorenz-63 system is a classic chaotic model defined by:

$$\begin{aligned}\frac{dx}{dt} &= \sigma(y - x) \\ \frac{dy}{dt} &= x(\rho - z) - y \\ \frac{dz}{dt} &= xy - \beta z\end{aligned}$$

Typical parameters:
 $\sigma = 10$, $\rho = 28$, $\beta = 8/3$

References

- [1] Andrew J Majda and John Harlim, *Filtering complex turbulent systems*, Cambridge University Press, 2012.